Plasmonic-Enhanced Laser-Induced Sonofusion: Overcoming the Temperature Barrier via Resonant Nanoparticle Heating

Consolidated Final Scientific Report

Dimitri Bolt

Academic supervisor: Prof. Dr. Ildar R. Gabitov

Mini-Research 1–8 | 31 October 2026

For preliminary scientific discussion

Abstract

This report develops a complete account of a reduced mathematical study of acoustic bubble compression and an additional laser energy channel mediated by gold nanoparticles. The three objectives are to construct a baseline bubble model, extend it by a physically ordered laser–gold–liquid–gas pathway, and determine whether ordinary or physically admissible extreme conditions support temperatures relevant to thermonuclear reactions. Rayleigh–Plesset and Keller–Miksis mechanics are combined progressively with gas energy, temperature-dependent heat capacity, a fixed water-vapor inventory, Mie absorption, resolved stationary heat transfer and a moving radial thermal model.

For a 6 μm initial bubble in water at 30 kHz, one 20 nm-radius gold particle prescribed to remain 100 nm from the wall, and a 532 nm, 10 ns, 10 J/m^2 pulse, the dynamically coupled model predicts global peak increments of 1.408816 and 2.184879 μK at acoustic amplitudes 1.00 and 1.15 bar. The first is conditional on the reduced geometry and transport; the second is a numerically complete but physically formal fixed-composition scenario. Wet 1.30 bar trajectories stop at the 6000 K water-coefficient boundary, so their global peak increment is undefined. Dry formal continuations reach approximately 6264 K for air and 22809 K for ground-state argon. Much larger historical RP estimates are retained as diagnostics of invalid extrapolation.

New bounded Gilmore–Tait calculations resolve liquid-model sensitivity without importing temperatures from another trajectory. New caloric and energy audits distinguish absorption, storage, signed gas heat and compression work. The specified single-particle pathway produces a very small peak effect, but this is not a universal efficiency or temperature limit. The program establishes neither a physically reliable maximum temperature nor thermonuclear feasibility. A local hot region, ion/electron temperatures, fuel inventory and reaction yield remain undetermined. The project title states the research hypothesis; overcoming the barrier has not been demonstrated.

Contents

1 Introduction, motivation and research objectives **3**

2 Overview of the mathematical modeling framework **4**

3 Acoustic bubble dynamics and compressibility **5**

4 Gas thermodynamics and the temperature barrier **8**

5 Plasmonic absorption and gold nanoparticle heating **10**

6 AuNP–bubble energy transfer **12**

7 Integrated laser–AuNP–bubble model **14**

8 Timing and useful energy delivery **16**

9 Energy analysis and idealized concentration **17**

10 Representative extreme scenarios and scientific status **19**

11 Implications for thermonuclear reactions **20**

12 Discussion and conclusions **22**

References **23**

Guide to the Supporting Mini-Research Reports **24**

How to read this report. The model, reference parameters, principal results and limitations are stated here. Footnotes identify exact supporting reports and verified section titles; they are routes to derivation and numerical evidence, not prerequisites for understanding the conclusions. Literature references are listed separately. Unless indicated otherwise, temperature means the spatially uniform gas temperature of a specified model. A completed calculation is not an experimental validation.

1 Introduction, motivation and research objectives

Acoustic rarefaction expands a gas bubble and subsequent compression can concentrate mechanical energy into a short interval. The scientific question is whether additional optical absorption by gold nanoparticles can deliver useful energy to the gas before its temperature maximum, and whether the resulting conditions have any defensible connection to thermonuclear reactions. These questions require energy accounting and model applicability as well as a large computed temperature.

The program has three objectives:

1. **Baseline mathematical model of sonofusion-related bubble dynamics.** Determine radius, wall speed, pressure and gas temperature under acoustic excitation without laser heating or nanoparticles.
2. **Laser-enhanced model with gold nanoparticles.** Add the ordered pathway Laser → Au absorption → metal heating → surrounding liquid → bubble boundary → gas energy and mechanical response; quantify the change in the gas-temperature maximum.
3. **Feasibility of extreme temperatures relevant to thermonuclear reactions.** Examine ordinary and rare extreme configurations without assuming a favorable or unfavorable result in advance, and distinguish model temperature from the state required to calculate nuclear reactions.

The first objective is achieved as a hierarchy of deterministic spherical models. The second is achieved within a conservative moving radial reduction with prescribed particle placement. The third has a bounded, unresolved outcome: the available models expose limiting mechanisms and invalid extrapolations but do not establish a physically attainable upper temperature, plasma hot-region state or nuclear yield.

The sequence has a specific logic. Mini-Research 1 supplies incompressible radius dynamics; Mini-Research 2 adds first-order liquid compressibility. Mini-Research 3 distinguishes adiabatic reconstruction, one-way thermal post-processing and coupled gas mechanics. Mini-Research 4 introduces variable species heat capacities and a consistent fixed vapor inventory, while assessing missing chemistry. Mini-Research 5 determines optical absorption and heat release by isolated gold spheres. Mini-Research 6 resolves a finite particle and stationary bubble, then tests the applicability of that frozen geometry. Mini-Research 7 couples mechanics, gas, moving water and gold storage. Mini-Research 8 evaluates the physical meaning and remaining uncertainty of the whole chain.

Sonoluminescence observations motivate this work, but their thermometry and host liquids must be identified. Flannigan and Suslick observed emission consistent with plasma formation in concentrated aqueous sulfuric acid [12]. This does not calibrate a uniform water-bubble temperature, identify its ion temperature or demonstrate fusion. The deuterated-acetone calculations of Sadighi-Bonabi and Gheshlaghi [11] use substantially different inventories, conditions and closures. Neither source supplies a matched experimental validation of the present water/AuNP model.

2 Overview of the mathematical modeling framework

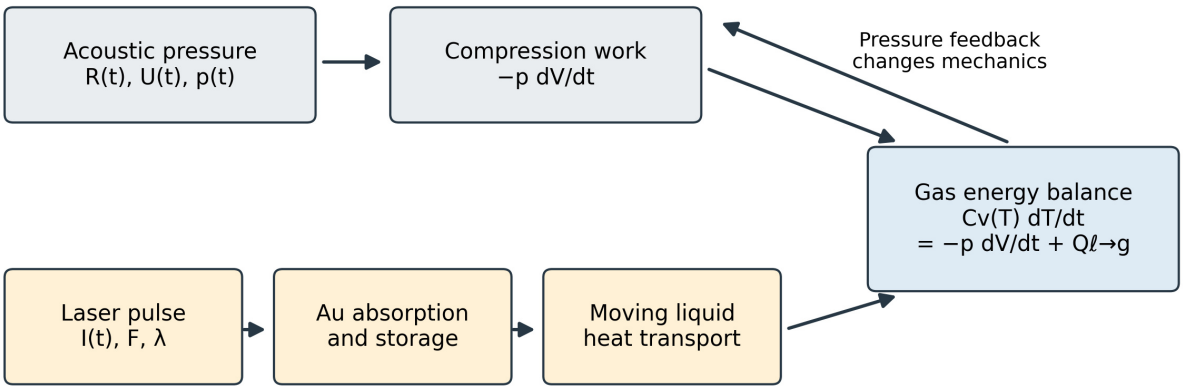


Figure 1: The acoustic and optical branches meet in the gas-energy equation. Gas pressure feeds back on radius dynamics. The laser is not represented as direct gas heating. The gold source in the moving model is a spherical average of a finite, localized particle.

The reference bubble starts at rest in unbounded water. The applied pressure is specified locally in time,

$$p_\infty(t) = p_0 - P_a \sin(2\pi f t), \quad R(0) = R_0, \quad U(0) = 0, \quad T_g(0) = T_0. \tag{1}$$

The first quarter-cycle is rarefaction; P_a is peak acoustic amplitude, not absolute pressure. The usual requested observation interval is five periods, $I = [0, 5/f]$; the new mechanical comparison uses one complete period.

Quantity	Value	Quantity	Value
R_0	6 μm	f	30 kHz
T_0	293.15 K	p_0	101325 Pa
Mechanical ρ_0	998.2 kg/m^3	c_0	1482 m/s
μ	0.001002 Pa s	σ	0.07274 N/m
Initial p_v	2339 Pa	$5/f$	166.667 μs
Thermal $\rho_\ell c_\ell$	4.187×10^6 J/(m^3 K)	k_ℓ	0.6 W/(m K)
Air reference k_g	0.0264 W/(m K)	Ar reference k_g	0.0177 W/(m K)

The mechanical water properties are rounded near-ambient references; thermal water properties follow the separate photothermal parameter set. Keeping that small distinction explicit avoids interpreting numerical consistency as a new property measurement. Gas conductivities are held at their reference values and are not high-temperature mixture transport laws.

The state and energy conventions change between early exploratory models and the final coupled model. The final wet gas is a single fixed mixture with initial total pressure $p_0 + 2\sigma/R_0$. Water vapor is included in its moles, pressure and energy. There is no additional constant vapor-pressure term in that formulation. Earlier models with a separately added p_v are identified as such below.

3 Acoustic bubble dynamics and compressibility

3.1 Rayleigh–Plesset baseline

For an incompressible Newtonian liquid, spherical momentum balance gives

$$\rho_0(R\dot{U} + \tfrac{3}{2}U^2) = p_d(R) + p_v - p_\infty - 2\sigma/R - 4\mu U/R, \quad U = \dot{R}. \tag{2}$$

The initial exploratory closure is $p_d = p_{d0}(R_0/R)^{3\kappa}$, with $\kappa = 1.4$ and $p_{d0} = p_0 + 2\sigma/R_0 - p_v$. Here p_d is dry-gas partial pressure. Static balance fixes the normalization, rather than an arbitrary maximum-to-minimum volume ratio. This is the Mini-Research 1 baseline.[1]

At 1.0 bar, the five-cycle radius range is approximately 4.12–11.34 μm and peak wall speed about 7.74 m/s. At 1.3 bar, extreme supersonic speeds invalidate incompressibility. Converged five-cycle integration establishes neither physical applicability nor attractor/shape stability.

3.2 Keller–Miksis equation and pressure feedback

The first-order compressibility convention used throughout the program is

$$(1 - U/c_0)R\dot{U} + \tfrac{3}{2}(1 - U/(3c_0))U^2 = \frac{1 + U/c_0}{\rho_0}\Delta p + \frac{R}{\rho_0 c_0}\dot{\Delta p}, \tag{3}$$

where $\Delta p = p_g + p_c - p_\infty - 2\sigma/R - 4\mu U/R$. The final mixture uses $p_c = 0$; the early dry-gas closure uses $p_c = p_v$. Differentiating the viscous term contributes $-4\mu\dot{U}/R$ and gives the explicit denominator

$$D = R(1 - U/c_0) + 4\mu/(\rho_0 c_0). \tag{4}$$

For the polytrope $\dot{p}_g = -3\kappa p_g U/R$. For the energy-coupled fixed mixture, $\dot{p}_g = p_g(\dot{T}_g/T_g - 3U/R)$. The latter must be used when gas heating feeds back mechanically.[2]

The original incident field includes radial propagation [1]; Eq. (3) instead fixes the local forcing convention. Its pressure prefactor differs from Eq. (1) of Ref. [11]; the conventions are not mixed. The $c_0 \to \infty$ limit recovers RP. Denominator regularity, including at inward $U = -c_0$, does not establish physical validity.

[1] Mini-Research 1, "Mathematical Modeling of Single-Bubble Dynamics under Acoustic Excitation," Section "Mathematical model," `mini_research_01.pdf`.

[2] Mini-Research 2, "Compressible Bubble Dynamics and Extreme Collapse Conditions," Section "Keller–Miksis convention and explicit acceleration," `02_Compressible_Bubble_Dynamics.pdf`.

3.3 What strong collapse requires from the liquid model

Useful diagnostics are $M_0 = \max |U|/c_0$ and

$$\chi = \max \frac{|p_L - p_0|}{\rho_0 c_0^2}, \quad p_L = p_g + p_c - 2\sigma/R - 4\mu U/R. \tag{5}$$

For the strongest five-cycle MR2 polytropic KM case, $R_{\min} = 0.550809\,\mu$m, peak inward speed is 1072.98 m/s, $M_0 = 0.7240$ and $\chi = 1.276$. These are converged equation outputs at 1.3 bar, with strong liquid-compressibility warnings.[3]

KM resolves neither shock profiles nor shape modes/jets. Large χ motivates variable density and sound speed without predicting density change. Stability needs perturbations and geometry; a deterministic radius sweep supplies no rare-event probabilities.[4]

At 1.15 and 1.30 bar, minimum far-field absolute pressures are -13675 and -28675 Pa. Such liquid tension may be metastable. Missing nucleation, purity, geometry and exposure data prevent either rejecting it outright or establishing isolated-bubble operation.

3.4 New bounded Gilmore–Tait comparison

Mini-Research 8 isolates liquid-model sensitivity over $[0, 1/f]$ at 1.00, 1.15 and 1.30 bar, using identical initial conditions, MR2 polytropic gas and constant p_v. These mechanical comparisons do not replace MR7 thermodynamics.

For a barotropic Tait liquid,

$$\rho(p) = \rho_0 \left(\frac{p + B}{p_0 + B}\right)^{1/n}, \quad C_L^2 = \frac{n(p_L + B)}{\rho_L}, \quad H = \frac{n}{n-1}\left[\frac{p_L + B}{\rho_L} - \frac{p_\infty + B}{\rho_\infty}\right]. \tag{6}$$

Using the water exponent $n = 7.15$ [2], $B = \rho_0 c_0^2/n - p_0 = 3.06523 \times 10^8$ Pa matches the existing ambient sound speed without a collapse fit. The checked $p + B > 0$ branch supports this mechanical comparison, not liquid thermometry, tensile stability or phase behavior.

[3] Mini-Research 2, "Compressible Bubble Dynamics and Extreme Collapse Conditions," Section "Extreme-collapse analysis and event correspondence," `02_Compressible_Bubble_Dynamics.pdf`.

[4] Mini-Research 2, "Compressible Bubble Dynamics and Extreme Collapse Conditions," Section "Physical applicability and unresolved mechanisms," `02_Compressible_Bubble_Dynamics.pdf`.

The Gilmore equation used for the new comparison is

$$(1 - U/C_L)R\dot{U} + \tfrac{3}{2}(1 - U/(3C_L))U^2 = (1 + U/C_L)H + (1 - U/C_L)\frac{R}{C_L}\dot{H}, \tag{7}$$

where $\dot{H} = \dot{p}_L/\rho_L - \dot{p}_\infty/\rho_\infty$, including the full viscous acceleration derivative. Equilibrium, enthalpy, the implicit equation and the incompressible limit are independently checked; the explicit derivation is in the MR8 Technical Report.

Table 1: New one-cycle mechanical extrema. Temperature is not calculated on these trajectories. M_0 uses the common ambient sound speed, and χ the common ambient bulk-modulus scale.

P_a (bar)	Model	$R_{\min}$ (μm)	t_R (μs)	$\lvert U_- \rvert$ (m/s)	M_0	χ
1.00	KM	4.393090	20.092624	6.565	0.0044	0.0001
1.00	Gilmore	4.393164	20.092625	6.564	0.0044	0.0001
1.15	KM	1.228160	15.664319	154.233	0.1041	0.0439
1.15	Gilmore	1.233152	15.664393	152.478	0.1029	0.0431
1.30	KM	0.550809	18.341064	1072.981	0.7240	1.2758
1.30	Gilmore	0.513361	18.341191	1031.292	0.6959	1.7149

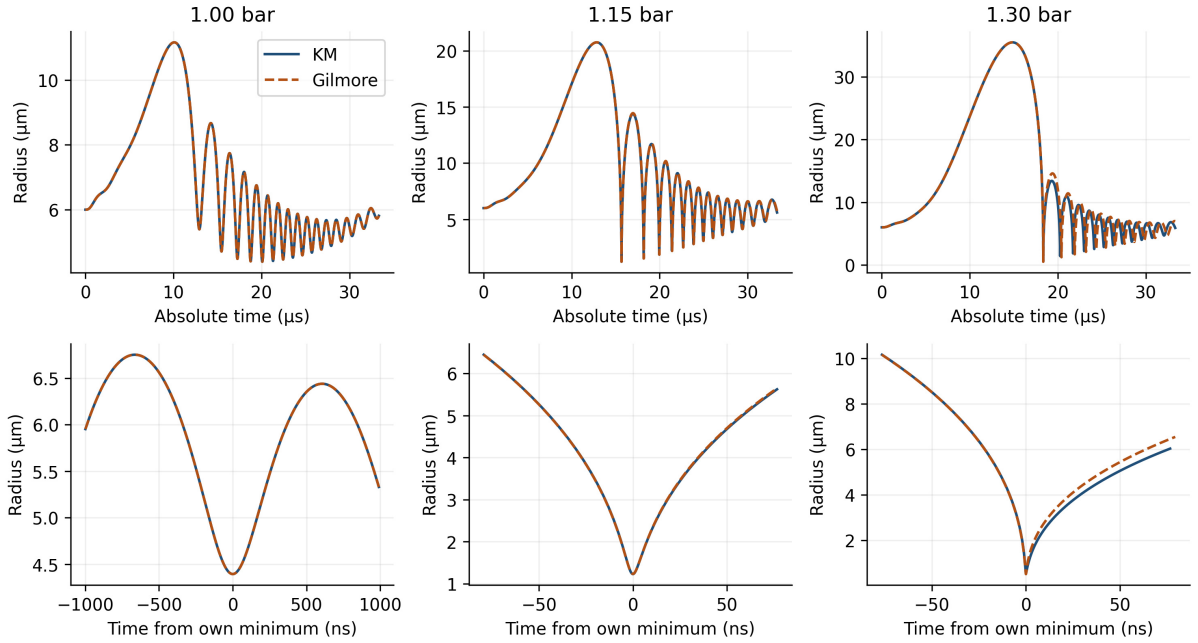


Figure 2: New mechanical comparison. Top: common absolute time axes. Bottom: each curve is centered on its own first-cycle deepest minimum; centering does not erase the difference in absolute event times. The 1.0 bar panel has a wider zoom.

At 1.30 bar, Gilmore reduces $R_{\min}$ by about 6.80% and inward peak speed by 3.89%. Its wall pressure reaches about 3.76 GPa, Tait density ratio 1.436 and local wall Mach 0.504. These are mechanical sensitivity diagnostics, not validated extreme states; no gas temperature is transferred between trajectories.

DOP853 refinement and independent Radau differ by at most 4.27×10^{-10} in minimum radius and 3.70×10^{-9} in peak speed, much less than the liquid-model spread. These empirical numerical checks do not validate reacting-gas/compressible-liquid physics.

4 Gas thermodynamics and the temperature barrier

4.1 Compression, signed heat and mechanical feedback

For a closed gas with no reaction or mass exchange,

$$\dot{U}_g = -p_g \dot{V} + Q_{\ell\to g}, \quad V = 4\pi R^3/3, \quad \Delta U_g = -\int p_g \, \mathrm{d}V + \int Q_{\ell\to g} \, \mathrm{d}t. \tag{8}$$

With constant molar c_V and no heat transfer, integration gives

$$T_g = T_0 (R_0/R)^{3(\gamma-1)}, \quad \gamma = 1 + R_u/c_V. \tag{9}$$

The normalization uses the initial state. Replacing R_0 by an expansion maximum without changing the thermodynamic reference would be incorrect.

Mini-Research 3 compares adiabatic reconstruction, one-way heat transfer on prescribed mechanics and coupled gas–KM evolution. Its noncondensable-gas energy excludes the separately prescribed mechanical vapor pressure; its balance is not a full vapor/liquid budget.[5] The closure $Q_{\ell\to g} = 4\pi\eta k_g R(T_0 - T_g)$, nominal $\eta = 1$, heats cold gas and cools hot gas. Coupling changes pressure and compression work, rather than simply subtracting heat from a fixed peak.[6]

P_a (bar)	Adiabatic KM (K)	One-way thermal (K)	Coupled thermal (K)
1.00	426.14	441.26	707.13
1.15	1966.82	3898.73	4423.44
1.30	5148.38	15174.00	10555.02

These numerical MR3 values use $\gamma = 1.4$ and the early vapor convention. They are neither laser effects nor measurements, and cannot replace later mixture results.

Temperature extrema are found independently: at $U = 0$, hot gas losing heat has $\dot{T}_g < 0$, so its nearby thermal peak is earlier. Global radius/temperature extrema can belong to different cycles; their time difference need not describe corresponding collapses.

4.2 Variable heat capacity and the fixed vapor inventory

The later closed mixture has fixed species amounts n_i, $n = \sum_i n_i$, and

$$p_g = nR_uT_g/V, \qquad C_V(T_g) = \sum_i n_i[c_{P,i}(T_g) - R_u], \tag{10}$$

$$\Delta U_g(T_g) = \sum_i n_i \int_{T_0}^{T_g} [c_{P,i}(T) - R_u] \, \mathrm{d}T, \qquad \dot{T}_g = \frac{-p_g\dot{V} + Q_{\ell\to g}}{C_V(T_g)}. \tag{11}$$

NASA9 species heat capacities [4] are integrated on their validity intervals with continuous energy joins. A diagnostic variable $\gamma(T)$ inserted into $pV^{\gamma(T)} =$ constant would not solve this energy equation.[7]

Initially $n_v = p_vV_0/(R_uT_0)$ and $n_d = (p_0 + 2\sigma/R_0 - p_v)V_0/(R_uT_0)$: about 1.86% vapor, with 79/21 N_2/O_2 or ground-state Ar as dry gas. The same inventory supplies pressure and heat capacity, with no second p_v. Fixed-total-pressure vapor contrasts also change dry moles.

[5] Mini-Research 3, "In-Bubble Thermodynamics: Compression Heating, Energy Balance, and Extreme Temperature Estimates," Section "Gas energy, signed heat transfer and vapor accounting," `03_In_Bubble_Thermodynamics.pdf`.

[6] Mini-Research 3, "In-Bubble Thermodynamics: Compression Heating, Energy Balance, and Extreme Temperature Estimates," Section "One-way and fully coupled formulations," `03_In_Bubble_Thermodynamics.pdf`.

[7] Mini-Research 4, "Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization," Section "Thermodynamically consistent formulation," `04_Temperature_Barrier.pdf`.

The coupled MR4 saturated-air peaks are about 552.9 and 2798.4 K at 1.00 and 1.15 bar, versus the earlier 707.1 and 4423.4 K. This contrast combines vapor accounting and caloric response. Wet 1.30 bar stops at its coefficient boundary and has no complete global maximum.

Fixed-species rotational/vibrational heat capacity does not describe evaporation/condensation enthalpy, dissociation or ionization. Those processes change mass, species populations or charge/energy storage. Vapor trapping is a supported mechanism [5]; quantifying it here requires mass transfer and chemistry.

The H_2O coefficients cover 200–6000 K and N_2/O_2 200–20000 K. Termination prevents extrapolation, not higher physical temperatures. Analytic ground-state Ar has no polynomial ceiling but omits excitation/ionization.

4.3 High-temperature interpretation and retained formal extrema

Above 6000 K, the archived wet closure cannot continue. Higher-temperature neutral coefficients alone would still omit composition changes. Dense-state nonideality, finite molecular volume and relaxation require separate assessment; evaluable polynomials do not validate the physical state.

For a reacting closed mixture one can write

$$U_g = \sum_i n_i u_i(T), \qquad C_V \dot{T} = -p_g \dot{V} + Q_{\ell \to g} - \sum_i u_i(T) \dot{n}_i, \tag{12}$$

with formation energies and consistent species balances. With mass exchange, interface enthalpy and species fluxes must also enter the energy and mass balances. For a consistent equilibrium closure $U(T, V)$, the corresponding equation is

$$\dot{T} = \frac{-[p + U_V]\dot{V} + Q_{\ell \to g}}{U_T}, \qquad \dot{p} = p_T \dot{T} + p_V \dot{V}. \tag{13}$$

Nonequilibrium ionization additionally requires electron/ion energy balances. Saha equilibrium needs nuclei/charge constraints, partition data and evidence of local thermodynamic equilibrium, none supplied by the scalar gas temperature.

Mini-Research 4 compares NIST dissociation/ionization costs with frozen sensible/work budgets as energy scales, not conversion fractions.[8] Ar first ionization costs 15.759610 eV [10]; dividing by k_B determines neither onset nor ionized fraction. Density, entropy, recombination and time remain essential.

The largest complete MR3 adiabatic RP estimate is 1.13046×10^6 K at 1.295 bar; the censored 1.16720×10^6 K at 1.185 bar has no global-maximum status. Both violate incompressible/gas-state assumptions. A one-way argon diagnostic reaches about 213182 K at 1.30 bar without self-consistent argon mechanical feedback.[9] The integrated dry-Ar value, 22808.7 K, is also formal. Section 12 separates completed extrema, censored observations and absent physical bounds; none measures a hot region.

[8] Mini-Research 4, "Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization," Section "Dissociation and ionization: quantitative assessment," `04_Temperature_Barrier.pdf`.

[9] Mini-Research 3, "In-Bubble Thermodynamics: Compression Heating, Energy Balance, and Extreme Temperature Estimates," Section "Applicability of extreme scenarios," `03_In_Bubble_Thermodynamics.pdf`.

5 Plasmonic absorption and gold nanoparticle heating

5.1 Optical input and the finite photon budget

For a local incident intensity $I(t)$, a sphere absorbs $P_{\mathrm{abs}} = \sigma_{\mathrm{abs}} I(t)$ and $E_{\mathrm{abs}} = \sigma_{\mathrm{abs}} F$, with fluence $F = \int I(t)\,\mathrm{d}t$. Scattering removes energy from the incident beam without directly heating the metal. Mie theory is used for quantitative cross sections, with room-temperature Johnson–Christy Au data and homogeneous real host index $n_m = 1.33$ [6].[10]

The small-sphere limit has normalized polarizability

$$\alpha_0 = 4\pi a^3 \frac{\varepsilon_{\mathrm{Au}} - \varepsilon_m}{\varepsilon_{\mathrm{Au}} + 2\varepsilon_m}, \quad \alpha = \frac{\alpha_0}{1 - i k_m^3 \alpha_0/(6\pi)}, \quad k_m = 2\pi n_m/\lambda_0. \tag{14}$$

For the $e^{-i\omega t}$ convention and passive $n + i\kappa$, $\sigma_{\mathrm{ext}} = k_m \,\mathrm{Im}\,\alpha$, $\sigma_{\mathrm{sca}} = k_m^4 |\alpha|^2/(6\pi)$ and $\sigma_{\mathrm{abs}} = \sigma_{\mathrm{ext}} - \sigma_{\mathrm{sca}} \geq 0$. Radiation correction enforces the dipole optical theorem; higher multipoles require the full Mie series, checked for convergence and against an independent implementation.

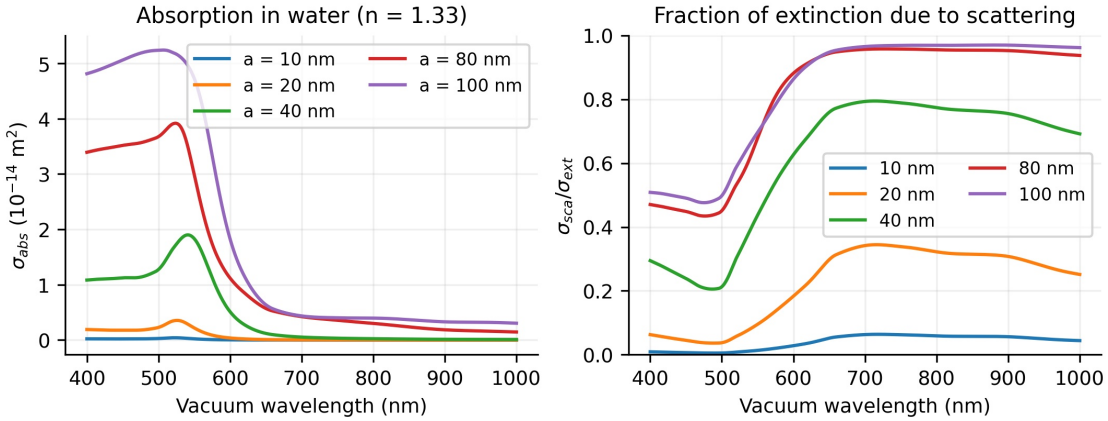


Figure 3: Unmodified MR5 figure. Absorption and scattering vary differently with particle radius. Discrete spectral maxima are properties of the declared cold-material optical model.

At $a = 20$ nm and 532 nm, $\sigma_{\mathrm{abs}} = 3.43373245 \times 10^{-15}$ m^2 gives 34.3373 fJ at $F = 10$ J/m^2. The corresponding 100 μm-radius flat beam contains 314.159 nJ; that full beam energy is not assigned to the particle or gas.

[10] Mini-Research 5, "Plasmonic Absorption and Photothermal Heating of Gold Nanoparticles," Section "Optical theorem, Mie theory and spectral results," `05_Plasmonic_Heating.pdf`.

5.2 Thermal storage, transfer and material feasibility

For a uniform metal lattice at adequately long times,

$$C_{\mathrm{Au}}\dot{T}_{\mathrm{Au}} = P_{\mathrm{abs}} - Q_{\mathrm{Au}\to\ell}, \qquad C_{\mathrm{Au}} = \tfrac{4}{3}\pi a^3 \rho_{\mathrm{Au}} c_{\mathrm{Au}}, \qquad Q_{\mathrm{Au}\to\ell} = \int_{S_{\mathrm{Au}}} G(T_{\mathrm{Au}} - T_\ell)\,\mathrm{d}A. \tag{15}$$

The reference coefficients are $\rho_{\mathrm{Au}} = 19320$ kg/m^3, $c_{\mathrm{Au}} = 129$ J/(kg K), $k_{\mathrm{Au}} = 317$ W/(m K), and nominal $G = 10^8$ W/(m^2 K), following the photothermal parameter set [7]. For $a = 20$ nm, $C_{\mathrm{Au}} = 8.352 \times 10^{-17}$ J/K. The 10 ns pulse exceeds the 1.7 ps electron–phonon and 3.15 ps internal diffusion scales, an equilibrium screen that does not validate constant optics.

The isolated MR5 sphere at $F = 1$ J/m^2 gives Au/liquid-surface rises of 2.319/1.641 K; linear $F = 10$ gives 23.19/16.41 K. MR7 instead gives about 6.81 K Au rise. This discrepancy prevents local material certification by a spherical average; it is not a calibrated moving-model error factor.[11]

MR5's moderate design permits about 3.182 pJ for a 100 nm-radius sphere at 506 nm, 100 ns and a chosen 30 K rise screen. Its macroscopic layer absorbs 235.051 μJ from 0.785398 mJ, without a mechanism concentrating this energy into one bubble. Neither design maximum is universal.[12]

Bulk Au melting at 1337.33 K corresponds to about 87.2 fJ sensible energy for a 20 nm particle before latent heat/losses [9]. Size, coating, reshaping, phase change and temperature-dependent optics prevent treating this diagnostic as a universal damage threshold.

Au-adjacent water may nucleate another cavity, altering optics and contact. Kitz et al.'s roughly 600 J/m^2 threshold at 532 nm for supported 90 nm-diameter particles and 5 ns pulses [8] does not calibrate this moving 40 nm-diameter particle. Section 11.3 identifies the experimental-validation boundary.

[11] Mini-Research 5, "Plasmonic Absorption and Photothermal Heating of Gold Nanoparticles," Section "Continuous and pulsed heat transfer," `05_Plasmonic_Heating.pdf`.

[12] Mini-Research 5, "Plasmonic Absorption and Photothermal Heating of Gold Nanoparticles," Section "Extreme regimes, admissible candidates and formal continuation," `05_Plasmonic_Heating.pdf`.

6 AuNP–bubble energy transfer

6.1 Resolved frozen geometry: what it measures

Mini-Research 6 resolves a finite Au sphere, liquid diffusion and a stationary spherical gas boundary in an axisymmetric meridian. Its reference has bubble radius 6 μm, $a = 20$ nm, surface gap $d = 100$ nm, 532 nm, 10 ns and $F = 1$ J/m^2. The tagged laser perturbation starts at zero; the gas reference temperature is held fixed.[13]

The physical gas-boundary law is $q_{\ell g} = h_g(T_{\ell,w} - T_g)$, positive into gas. For the stationary laser perturbation, $\delta q_{\ell g} = h_g \theta_{\ell,w}$ because the reference gas perturbation is zero. Summing the discretized heat equations yields

$$E_{\text{abs}} = E_{\text{Au}} + E_\ell + Q_g + Q_{\text{bath}}. \tag{16}$$

Internal fluxes cancel between subsystems; bath escape is independently integrated.

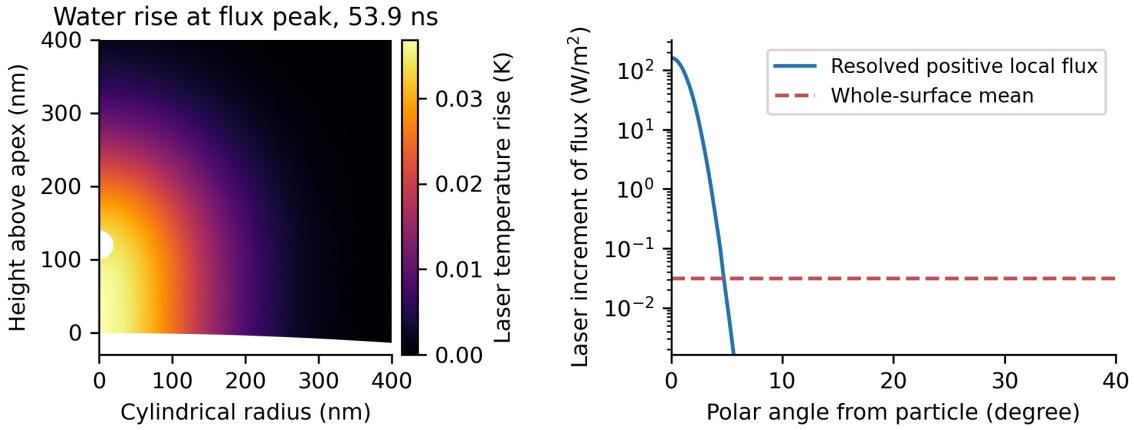


Figure 4: Unmodified MR6 figure. The finite-particle temperature and facing bubble-wall heat flux are strongly localized. These local fields are absent from an angularly smeared source.

Only 0.3258% reaches gas by 2 μs (11.1865 aJ of 3.43373 fJ); gas power peaks about 53.94 ns after pulse start. Flux is strongly localized: maximum/mean is about 5197, with 90% on 0.0481% of the surface. These stationary observables do not establish collapse efficiency.

The ideal point-source fraction $(R/r)\,\text{erfc}[(r-R)/\sqrt{4\alpha_\ell t}]$ for an absorbing stationary sphere misses finite-Au near-contact geometry. It cannot replace moving-interface transport with a prescribed efficiency or delay.

[13] Mini-Research 6, "Frozen Geometry AuNP–Bubble Thermal Coupling," Section "Resolved thermal problem and the signed interface," `06_AuNP_Bubble_Coupling.pdf`.

6.2 Particle motion and the limits of freezing the geometry

In the incompressible radial field $u_\ell(r) = R^2U/r^2$, a passive point center satisfies

$$r_{\mathrm{np}}^3 - R^3 = \text{constant}, \qquad d = r_{\mathrm{np}} - R - a. \tag{17}$$

Thus $\dot{d} = [(R/r_{\mathrm{np}})^2 - 1]U$: passive separation grows during collapse and contact may occur during expansion. Fixed centers and prescribed constant gaps are different assumptions; no retaining mechanism is derived.[14]

MR8 evaluates the slip required by the prescribed constant-gap law relative to the MR7 incompressible liquid velocity at the particle center:

$$v_{\mathrm{slip}} = U\left[1 - (R/(R+a+d))^2\right], \quad \mathrm{Re}_{\mathrm{np}} = 2a\rho_0|v_{\mathrm{slip}}|/\mu. \tag{18}$$

P_a (bar)	Required slip (m/s)	Particle Reynolds number
1.00	0.42084	0.01677
1.15	24.437	0.97377
1.30	220.64	8.7922

Required slip reaches about 24.4 m/s at 1.15 bar and 221 m/s before wet 1.30 bar censoring. These are imposed kinematic demands, not predicted motion. The 1.714 ns Stokes reference cannot justify creeping-flow retention at particle Reynolds number 8.79; the background incompressible flow is also conditional.

Frozen geometry requires small displacement throughout diffusion, screened by $|U|L/\alpha_\ell$ and radius/gap changes. MR6's late strong-collapse applications fail these screens despite convergence of the stationary heat problem.

Multiple-particle superposition requires separation and photon/interaction allowances. Local count differs from bulk concentration; local fluence already includes delivery losses. Near-contact optics remain outside homogeneous-host Mie theory.[15]

[14] Mini-Research 6, "Frozen Geometry AuNP–Bubble Thermal Coupling," Section "Kinematics, event resolution and applicability," `06_AuNP_Bubble_Coupling.pdf`.

[15] Mini-Research 6, "Frozen Geometry AuNP–Bubble Thermal Coupling," Section "Local energy, multiple particles and the reusable channel," `06_AuNP_Bubble_Coupling.pdf`.

7 Integrated laser–AuNP–bubble model

7.1 Moving water and one signed gas interface

Mini-Research 7 evolves R, U, T_g, temperatures in material water cells, Au storage and energy ledgers. Its moving coordinate is

$$\xi = (r^3 - R^3)/3, \qquad r = (R^3 + 3\xi)^{1/3}, \qquad \mathrm{d}V_\ell = 4\pi \mathrm{d}\xi. \tag{19}$$

Since $\mathrm{d}\xi/\mathrm{d}t = 0$, the constant-property liquid equation becomes

$$\rho_\ell c_\ell \partial_t T_\ell|_\xi = \partial_\xi(k_\ell r^4 \partial_\xi T_\ell) + q_{\mathrm{Au}}. \tag{20}$$

Positive constant capacities $C_i = 4\pi\rho_\ell c_\ell \Delta\xi_i$ and current spherical thermal resistances conserve energy within the incompressible approximation, not a compressible high-Mach flow model.

The gas conductance is $H_g = 4\pi\eta k_g R$ with nominal $\eta = 1$. Including liquid half-cell resistance K_w^{-1} gives

$$Q_{\ell\to g} = \frac{T_1 - T_g}{H_g^{-1} + K_w^{-1}}, \qquad T_{\ell,w} = T_g + Q_{\ell\to g}/H_g. \tag{21}$$

This current-state power enters gas and liquid with opposite signs. It replaces the earlier clamped-water heat loss; the old loss and a frozen MR6 flux are not added again. Gas energy, NASA heat capacity and its pressure derivative close the KM dynamics.[16]

The angularly averaged Au source uses positive weights summing to one for both sampled $\overline{T}_\ell = \sum_i w_i T_i$ and deposited heat. It omits finite-Au exclusion and local spreading resistance. Static integrated-delivery differences are about 0.04–1.68%, but radial Au rises are only 21–30% of resolved values. This agreement cannot validate local temperatures or moving near-contact error bounds.

Au storage follows the Mie input and its energy equation above. The prescribed 100 nm gap has no independent trapping model; peak increments remain conditional, with stronger formal limitations during severe collapse.

[16] Mini-Research 7, "Integrated Laser–AuNP–Bubble Model: Self-Consistent Dynamics and Peak Gas Temperature," Section "One signed interface, gas energy and mechanical feedback," `07_Integrated_Model.pdf`.

7.2 Quantitative effect of the nominal laser pulse

The principal comparison uses the same moving thermal model with and without a laser:

$$\Delta T_{\max} = \max_{t\in I} T_g^L(t) - \max_{t\in I} T_g^0(t). \tag{22}$$

Pairing a clamped-water baseline with a moving-water heated run would confound the laser effect with a model change. A paired global difference is undefined if either trajectory is censored. The source remains at the same absolute pulse time for both members; it is not recentered after the heated event shifts.[17]

Table 2: Nominal integrated comparison: $a = 20$ nm, prescribed $d = 100$ nm, 532 nm, 10 ns, $F = 10$ J/m^2; pulse center 5 μs before its selected no-laser radius reference. Global baseline temperatures are rounded; increments use cancellation-controlled pairs.

P_a (bar)	$T^0_{\max}$ (K)	$\Delta T_{\max}$ (μK)	Scientific interpretation
1.00	553.023	1.408816	Complete five cycles; conditional reduced model and unverified confinement
1.15	2798.281	2.184879	Complete five cycles; physically formal fixed composition and strong-collapse assumptions
1.30	undefined	undefined	Wet gas reaches its 6000 K coefficient boundary; only a bounded observed interval exists

Tiny increments require cancellation-controlled variational/nonlinear difference evolution; separate Radau pairs provide a check. Resolving the increment does not confer microkelvin accuracy on the absolute temperature or its physical model.

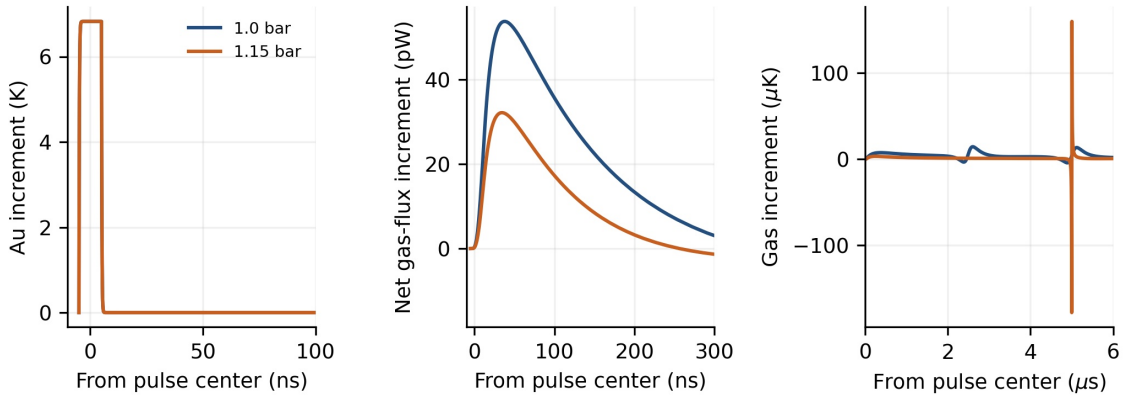


Figure 5: Unmodified MR7 figure. The paired source has a transport history and mechanical feedback. A transient temperature difference can exceed the difference of separately maximized temperatures when the trajectories shift.

At 1.00 bar mechanical feedback increases the one-way peak effect from about 0.858 to 1.409 μK. At 1.15 bar it decreases it from about 5.656 to 2.185 μK. Thus feedback has no universally favorable sign. Neither result is a universal limit on laser heating at other sizes, fluences, placements or gas states.

[17] Mini-Research 7, "Integrated Laser–AuNP–Bubble Model: Self-Consistent Dynamics and Peak Gas Temperature," Section "Paired temperatures and timing sensitivity," `07_Integrated_Model.pdf`.

8 Timing and useful energy delivery

The relevant question is how much energy reaches the gas while it can still influence the maximum. A large eventual transfer is insufficient if it arrives after that maximum; an earlier pulse can also act indirectly by changing pressure and compression work. Pulse duration, Au relaxation, water diffusion, interface transfer and hot-state lifetime must therefore be separated.

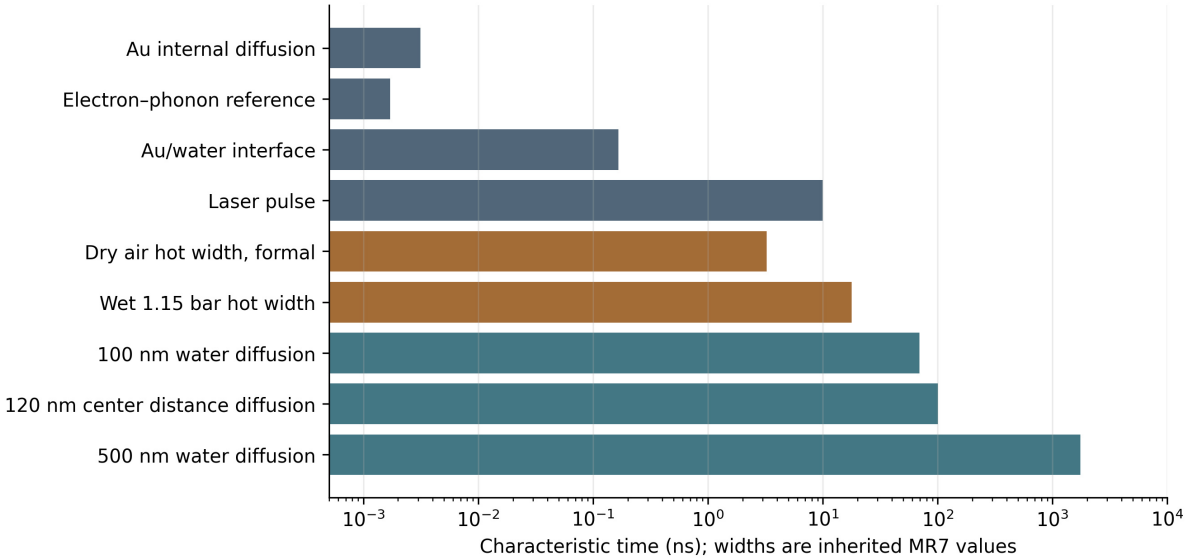


Figure 6: Reference scales from MR8 and inherited thermal widths from MR7. The width is the interval between half-height crossings relative to ambient temperature, not an emission lifetime, reaction-confinement time or universal collapse duration.

For the 100 nm surface gap, $d^2/\alpha_\ell = 69.8$ ns with $\alpha_\ell = 1.433 \times 10^{-7}$ m^2/s. A center-source distance $a + d = 120$ nm gives 100.5 ns. These are diffusion scales, not hard arrival times: the continuum heat equation has a small early tail. A 500 nm gap gives 1.745 μs, while a 1 nm gap formally gives 6.98 ps but raises continuum, interface and optical problems. Finite-particle surface geometry matters especially for such short distances.

For the nominal early MR7 pulse, the net gas-boundary power increment peaks about 38.6 ns after the pulse center at 1.00 bar and 34.9 ns at 1.15 bar within the declared 2 μs source window. The first-cycle thermal peaks occur about 2.459 and 5.000 μs after those centers. The net flux increment includes gas cooling and mechanics; it is not a pure tagged fraction of photons delivered to gas.

Inherited thermal half-height widths are about 406 ns at 1.00 bar, 17.86 ns at 1.15 bar, 3.232 ns for formal dry air at 1.30 bar and 7.030 ns for formal dry Ar. The censored wet 1.30 bar state has no complete width. Such reconstructed widths test temporal concentration, but do not measure chemical equilibration or establish the lifetime of an ion population.

The early 1.15 bar sensitivity is relatively stable under grid refinement. The event-centered late response changes by 89.6% between 48 and 96 liquid cells and remains unresolved. That late result cannot support a positive-pulse benefit or an optimum. The global nominal early-pulse result and the poorly resolved late timing response must remain distinct.

9 Energy analysis and idealized concentration

9.1 Conservation and the role of mechanical work

With signed gas power and an explicit bath, the thermal-subsystem equations are

$$\dot{U}_g = -p_g\dot{V} + Q_{\ell\to g}, \tag{23}$$

$$\dot{E}_\ell = Q_{\mathrm{Au}\to\ell} - Q_{\ell\to g} - Q_{\mathrm{bath}}, \tag{24}$$

$$\dot{E}_{\mathrm{Au}} = P_{\mathrm{abs}} - Q_{\mathrm{Au}\to\ell}, \tag{25}$$

$$\Delta(U_g + E_\ell + E_{\mathrm{Au}}) + W_{\mathrm{by}} + Q_{\mathrm{bath}} = E_{\mathrm{abs}}, \quad W_{\mathrm{by}} = \int p_g\dot{V}\,\mathrm{d}t. \tag{26}$$

Positive W_{by} means work by the gas; compression contributes negative work. Heat exchange changes sign and is never clipped. The identity covers the modeled thermal subsystem with mechanical work. It excludes liquid kinetic/acoustic energy, viscous heating and any work needed to retain a particle. It is not a total conservation statement for the externally driven apparatus.

Table 3: Inherited thermal energies independently reassembled in MR8, in pJ, at each run's endpoint. These include the acoustic background; they are not all laser-tagged energies. The 1.30* endpoint is censored at 18.5453 μs, whereas the other endpoints are 166.667 μs.

P_a (bar)	E_{abs}	E_{Au}	E_ℓ	ΔU_g	W_{by}	$Q_{\ell g}$
1.00	0.0343373	4.64855e-06	1157.09	-7.67078	-1148.83	-1156.5
1.15	0.0343373	1.4756e-05	3831.97	0.448837	-3842.28	-3841.83
1.30*	0.0343373	9.83682e-05	-210.908	7664.12	-7452.98	211.147

In all three cases the nominal local incident beam contains 314.159 nJ and a single particle absorbs 0.0343373 pJ. Au and liquid storage, gas internal-energy change, work, gas heat and bath escape are distinct. For the nominal complete 1.00 bar pair, final gas-energy change between members is about 4.50×10^{-19} J, net gas-heat change about -7.34×10^{-18} J and work-by-gas change about -7.79×10^{-18} J. Their relation is $\delta U_g = -\delta W_{\mathrm{by}} + \delta Q_{\ell g}$. The small negative heat increment does not contradict a positive peak-temperature change at an earlier time.

MR8 recomputes the saved gas/liquid/Au identities and independently integrates the nonlinear caloric gas increment from the saved paired temperatures. Maximum total residual normalized by the run's thermal scale is 1.35×10^{-6}. The independent paired gas residual is below 4.61×10^{-8} of absorption. Those checks verify the stated bookkeeping; they do not validate missing physical channels or infer a photon-delivery efficiency from net gas heat.

9.2 A favorable caloric allowance is not a universal bound

To isolate energy availability from transport, MR8 defines a deliberately idealized operation: deposit all of one particle's absorbed energy instantaneously into a fixed-composition, fixed-volume gas at its baseline peak, without losses or later work. The resulting temperature T_* solves

$$\int_{T_b}^{T_*} C_V(T)\,\mathrm{d}T = E_{\mathrm{abs}}. \tag{27}$$

The NASA integral is inverted only within the existing species domains. This is an upper allowance for that specified direct caloric operation, not an upper bound on a mechanically changing bubble or a local region of unknown mass.

Reference T (K)	Frozen gas	Ideal ΔT (K)
553.02	air wet	0.0334742
2798.28	air wet	0.0249898
6263.58	air dry	0.0234249
22808.72	argon dry	0.0590651

All rows use one 20 nm-radius particle at 532 nm and $F = 10$ J/m^2. Dry rows retain their already formal frozen gas assumptions. No wet 6000 K row is extended beyond its coefficient boundary. Actual Au storage, water diffusion and timing are intentionally absent from Eq. (27), while they remain present in MR7.

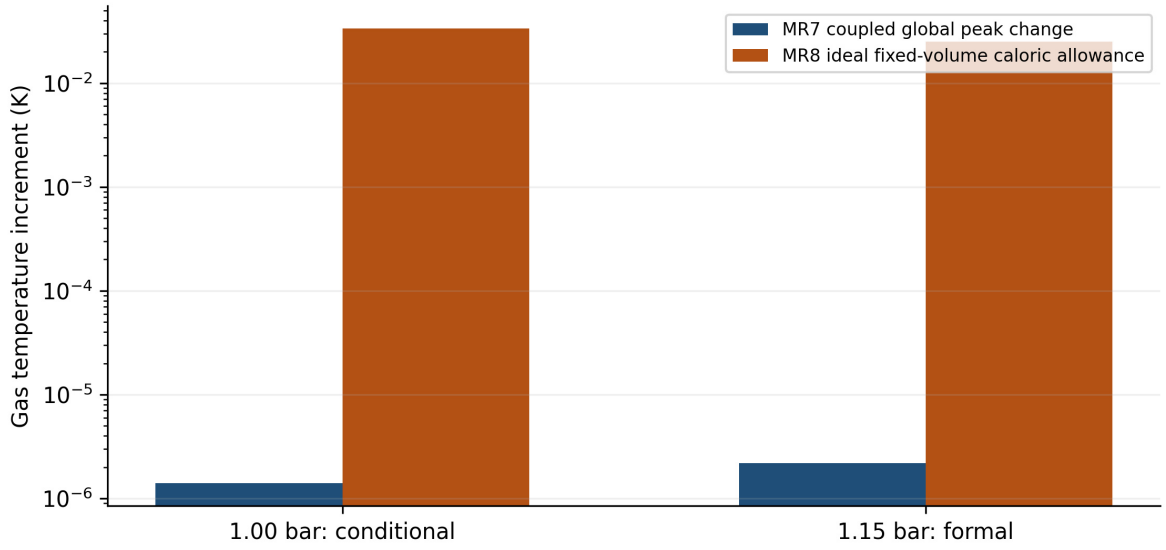


Figure 7: The ideal all-absorption caloric increments and the modeled coupled peak differences answer different questions. The large gap between them exposes the importance of transport, timing and changing compression; it is not a universal transfer-efficiency ratio.

The wet 1.00 and 1.15 bar allowances are about 0.03347 and 0.02499 K. They are much larger than the nominal microkelvin global changes, yet small compared with the corresponding baseline gas temperatures. Merely increasing a fluence in a cold linear optical model does not establish that the particle, liquid contact or gas composition remains intact.

No physically reliable upper $T_{\max}$ follows from this analysis. In the dynamic model the laser can change acoustic work; an energy-to-constant-capacity ratio ignores that feedback. In a local hot region the energy also depends on its unknown mass, geometry, composition and losses. The program supplies neither those data nor a probability model for rare energy-concentration events.

10 Representative extreme scenarios and scientific status

The categories used here have distinct meanings: **V**, validated within the declared mathematical model (equations and numerical checks); **C**, physically grounded conditional prediction; **F**, numerically formal extrapolation or extremum; and **U**, unresolved physical scenario. V does not mean experimentally validated. A scenario may have numerical status V and physical status F simultaneously.

Scenario and model	Temperature/result	Numerical status	Physical interpretation and missing evidence
Water, wet air, 1.00 bar, MR7 nominal pulse	$T_b \simeq 553.023$ K; $+1.408816\,\mu$K	V; five cycles complete	C; fixed gas, unverified retention, local Au field and gas conductance need assessment
Water, wet air, 1.15 bar, same pulse	$T_b \simeq 2798.281$ K; $+2.184879\,\mu$K	V; five cycles complete	F; tensile drive, 130 MPa gas pressure, chemistry/nonideal EOS and local material state unresolved
Water, wet air, 1.30 bar, MR7	Observed 6000 K boundary; global T and ΔT undefined	Censored interval; no completed peak	F/U; gas coefficients end; pressure about 2 GPa and density about 1157 kg/m^3 challenge ideal-gas/flow closures
Water, dry air, 1.30 bar, MR7 no laser	6263.58 K	V; complete	F; variable molecular capacity but fixed chemistry, density about 1261 kg/m^3; reacting nonideal gas needed
Water, dry ground-state Ar, 1.30 bar, MR7 no laser	22808.7 K	V; complete	F; missing excitation/ionization, nonideal gas and suitable transport
MR3 adiabatic RP, 1.295 bar	1.13046×10^6 K	Complete saved interval	F; incompressibility and gas closure grossly fail; not evidence of a realizable hot state
MR3 adiabatic RP, 1.185 bar	Observed 1.16720×10^6 K; global undefined	Censored at technical boundary	F; observed diagnostic only, no physical temperature cap or completed maximum
MR8 KM/Gilmore, 1.30 bar, common polytrope	$R_{\min}$: 0.550809 and 0.513361 μm; no T	V; one-cycle comparison	F; liquid-model sensitivity isolated; coupled thermodynamics and stability still required
Rare localized hot region; thermonuclear fuel	T_i, T_e, n_D, V_{hot}, τ unknown	No computed state	U; spatial reacting flow, fuel inventory and independently resolved diagnostics required

The initial radius, drive frequency and liquid are common where stated; the temperature rows retain their different gas and energy closures. The table is a finite representative selection, not a probability distribution or a ranking of all attainable conditions. High formal temperatures are preserved without being promoted into physical predictions or silently excluded.

11 Implications for thermonuclear reactions

11.1 What a temperature statement would have to specify

The uniform model temperature T_g summarizes a caloric state of a fixed mixture. It is not a prediction of the maximum in a spatial hot region. If the gas becomes a nonequilibrium plasma, ion and electron temperatures may differ; neither is determined by assigning the scalar T_g to a separate subsystem. Spectroscopic excitation, vibrational and continuum-fit temperatures are also distinct observables.

For identical D nuclei with a suitable local velocity distribution, the expected D–D reaction count would require

$$N_{\mathrm{DD}} = \int \frac{1}{2} n_D(\mathbf{x}, t)^2 \langle \sigma v \rangle_{\mathrm{DD}}(\mathbf{x}, t)\, \mathrm{d}V\, \mathrm{d}t. \tag{28}$$

The factor 1/2 avoids double counting. For a Maxwellian ion population, the reactivity is a function of ion temperature obtained from the physical nuclear cross section and relative-velocity distribution. Standard branch-specific parametrizations exist [14]; they cannot be applied with arbitrary density, temperature or duration. A hot volume must have a defined composition, density field and lifetime; a threshold temperature alone supplies no count or energy gain.

The present program specifies ordinary water and idealized air/Ar gas, not a deuterium-enriched fuel inventory. It supplies no isotope-resolved n_D, ion velocity distribution, T_i, T_e, spatial hot volume or reacting lifetime. A natural isotope abundance is not an adequate substitute for vapor capture, chemical composition and reacting-ion density during collapse. Therefore N_{DD} is undetermined; it is not assigned zero and is not estimated by inventing these missing quantities.

The very high formal RP temperatures cannot repair those omissions. Their underlying flow and EOS assumptions have already failed. Conversely, the modest or coefficient-censored temperatures in the current reduced model cannot prove all rare physical hot regions impossible. A universal negative conclusion would require a justified bound over the admissible physical configurations, which has not been obtained.

11.2 Water, acid experiments and deuterated acetone

Sadighi-Bonabi and Gheshlaghi's hydrochemical Table III lists 1.6×10^6 K for deuterated acetone at $-28.5\,^{\circ}$C with $R_0 = 8\,\mu$m, 38 kHz and 14.4 bar; their laser Table IV lists 1.9×10^6 K with $R_0 = 4.5\,\mu$m, 30 kHz and 1.5 bar [11]. Those entries change initial and driving conditions as well as model. Their plasma-core discussion invokes a hotter central region; it is not a measured core temperature in this water/AuNP system. They do not model the same nanoparticle-mediated transfer channel. No direct causal enhancement or nuclear yield is imported from that comparison.

11.3 Comparison with Experimental Evidence and Validation Status

Experimental measurements. Flannigan and Suslick measured atomic/molecular emission in concentrated aqueous sulfuric acid and interpreted excited and ionic emission as evidence consistent with plasma formation [12]. Their Ar emission thermometry includes a reported 15200 ± 1900 K at 2.8 bar. Xu and Suslick fitted OH rovibronic emission from a moving bubble in phosphoric acid, obtaining emission temperatures of 6200–9500 K over 1.9–3.1 bar [13]. These are emitter-specific spectroscopic temperatures under different host and driving conditions, not measurements of the entire gas, ion temperature or a central hot region. Kitz et al. measured vapor-bubble nucleation around supported 90 nm-diameter Au particles under 5 ns pulses, with a 532 nm threshold near 600 J/m^2 [8]; this probes a particle/phase-change observable, not acoustic gas heating or fusion.

Other numerical models and component comparisons. The acetone temperatures above are theoretical predictions, not experimental measurements or results of this program. Our Mie/conduction limits and stationary finite-Au comparisons test individual components under specified assumptions. Agreement in absorption, an isolated heating scale or integrated stationary heat delivery does not validate a complete moving laser–AuNP–gas system.

Own numerical checks and missing direct validation. Solver refinement, independent integrators and energy residuals test the implemented equations. They do not test whether those equations reproduce an experiment. No matched experiment supplies the full acoustic drive, radius history, gas/isotope composition, particle placement, optical input and local thermal observables needed to validate the complete coupled model. The program therefore has component-level consistency and numerical convergence evidence, but no direct experimental validation of its integrated peak-temperature prediction or DD yield.

12 Discussion and conclusions

12.1 Answers to the scientific questions

The spherical hierarchy couples acoustic mechanics and fixed-inventory gas thermodynamics to Mie Au absorption, metal storage and moving-water heat transport. Compression depends on liquid compressibility, gas capacity, vapor accounting and pressure feedback. Laser energy enters through water and the signed gas boundary.

Largest completed mathematical temperature. The largest completed value is approximately 1.13046×10^6 K in early adiabatic RP at 1.295 bar, with physically invalid incompressible/gas-state assumptions. The larger censored observation 1.16720×10^6 K is not a completed global maximum. MR7's dry-Ar 22808.7 K and dry-air 6263.58 K remain formal fixed-composition results. Wet 1.30 bar has no global temperature or increment after coefficient termination; 6000 K is not a ceiling. The 1.6–1.9 million K acetone values are other authors' theoretical predictions, neither our results nor experimental measurements.

Under the nominal single-particle conditions, the global peak increments remain 1.408816 and 2.184879 μK at 1.00 and 1.15 bar, respectively conditional and physically formal. None of these values establishes a universal physical maximum: invalid liquid/gas extrapolation, unresolved chemistry, phase change, local transport and spatial nonequilibrium prevent that inference.

MR8 demonstrates liquid-EOS sensitivity after convergence, separates ideal caloric allowance from useful dynamic heating, and audits signed heat and work. These checks do not validate omitted phase change, retention, chemistry or plasma temperatures.

No physically reliable universal upper temperature has been established. No justified claim of thermonuclear feasibility follows, including for rare extreme configurations; those configurations remain unresolved where the model lacks the necessary physics. This does not prove impossibility. DD yield needs spatial/temporal ion-temperature and velocity distributions, deuterium density, hot volume and lifetime, constrained by isotope/composition and reaction/background measurements. These inputs are absent; no DD yield or fusion energy gain is claimed.

12.2 Minimum evidence needed to resolve the uncertainty

The next discriminating work should be targeted rather than an unrestricted temperature search:

1. Measured drive/radius, composition, liquid state and nuclei, matched to coupled gas and compressible liquid dynamics; resolve shocks or shape modes where needed.
2. Consistent vapor/species/energy balances, finite-rate chemistry/ionization, nonideal pressure and transport; justify equilibrium against hot-state relaxation times.
3. Finite-Au moving transport with realizable retention, temperature-dependent optics, phase enthalpy and nucleation; independently test local Au temperature and gas-delivered heat.
4. Isotope-resolved fuel, ion/electron states and density–volume–time history, followed by justified reactivity and independent reaction/background diagnostics.

These are the specific missing tests needed for stronger physical conclusions, not additional results claimed by this report. The present consolidated account is suitable for preliminary scientific evaluation with those limits visible.

References

[1] J. B. Keller and M. Miksis, Bubble oscillations of large amplitude, *J. Acoust. Soc. Am.* **68**, 628–633 (1980). doi:10.1121/1.384720. Full journal reprint.

[2] F. Denner, The Gilmore–NASG model to predict single-bubble cavitation in compressible liquids, *Ultrason. Sonochem.* **70**, 105307 (2021). doi:10.1016/j.ultsonch.2020.105307; author manuscript, equations (10)–(25).

[3] D. Nagy and F. Hegedűs, Assessing the accuracy of the coupled-spherical-bubble approach for bubble pairs in an acoustic field, *Ultrason. Sonochem.* **123**, 107651 (2025). doi:10.1016/j.ultsonch.2025.107651. Full text.

[4] B. J. McBride, M. J. Zehe and S. Gordon, *NASA Glenn Coefficients for Calculating Thermodynamic Properties of Individual Species*, NASA/TP–2002–211556 (2002). NASA report; CEA species coefficients.

[5] R. Toegel, B. Gompf, R. Pecha and D. Lohse, Does Water Vapor Prevent Upscaling Sonoluminescence?, *Phys. Rev. Lett.* **85**, 3165–3168 (2000). doi:10.1103/PhysRevLett.85.3165.

[6] P. B. Johnson and R. W. Christy, Optical constants of the noble metals, *Phys. Rev. B* **6**, 4370–4379 (1972). doi:10.1103/PhysRevB.6.4370.

[7] G. Baffou and H. Rigneault, Femtosecond-pulsed optical heating of gold nanoparticles, *Phys. Rev. B* **84**, 035415 (2011). doi:10.1103/PhysRevB.84.035415. Author-hosted full text.

[8] M. Kitz, S. Preisser, A. Wetterwald, M. Jaeger, G. N. Thalmann and M. Frenz, Vapor bubble generation around gold nano-particles and its application to damaging of cells, *Biomed. Opt. Express* **2**, 291–304 (2011). doi:10.1364/BOE.2.000291; Full text.

[9] NIST, Kelvin: ITS-90, gold fixed point; NIST Chemistry WebBook, Au phase-change data, Stull vapor-pressure fit, 2142–3239 K.

[10] NIST Chemistry WebBook, water gas-phase thermochemistry; NIST Atomic Spectra Handbook, Ar ground states and ionization energy.

[11] R. Sadighi-Bonabi and M. Gheshlaghi, Laser induced sonofusion: A new road toward thermonuclear reactions, *AIP Adv.* **6**, 035218 (2016). doi:10.1063/1.4945343. Tables III–IV and the plasma-core discussion concern different models and host conditions.

[12] D. J. Flannigan and K. S. Suslick, Plasma formation and temperature measurement during single-bubble cavitation, *Nature* **434**, 52–55 (2005). doi:10.1038/nature03361.

[13] H. Xu and K. S. Suslick, Molecular Emission and Temperature Measurements from Single-Bubble Sonoluminescence, *Phys. Rev. Lett.* **104**, 244301 (2010). doi:10.1103/PhysRevLett.104.244301.

[14] UK Atomic Energy Authority, *PROCESS physics model: fusion reactions*, Bosch–Hale reactivity documentation and branch-specific parameters. Official scientific code documentation. Cites H.-S. Bosch and G. M. Hale, *Nucl. Fusion* **32**, 611–631 (1992), doi:10.1088/0029-5515/32/4/I07. No fit is evaluated here.

Guide to the Supporting Mini-Research Reports

No.	Scientific report and exact PDF	Content and useful sections
1	Mathematical Modeling of Single-Bubble Dynamics under Acoustic Excitation. mini_research_01.pdf	RP derivation, equilibrium, linear limit, radius/speed extrema. “Mathematical model”; “Collapse timing and numerical verification”; “Convergence and validity diagnostics.”
2	Compressible Bubble Dynamics and Extreme Collapse Conditions. 02_Compressible_Bubble_Dynamics.pdf	Full viscous KM convention, event correspondence, RP comparison and extreme applicability. “Keller–Miksis convention and explicit acceleration”; “Extreme-collapse analysis and event correspondence.”
3	In-Bubble Thermodynamics: Compression Heating, Energy Balance, and Extreme Temperature Estimates. 03_In_Bubble_Thermodynamics.pdf	Adiabatic, one-way and coupled gas energy. “Gas energy, signed heat transfer and vapor accounting”; “One-way and fully coupled formulations”; “Applicability of extreme scenarios.”
4	Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization. 04_Temperature_Barrier.pdf	NASA caloric law, fixed vapor inventory, coefficient domains and chemical energy scales. “Thermodynamically consistent formulation”; “Dissociation and ionization: quantitative assessment.”
5	Plasmonic Absorption and Photothermal Heating of Gold Nanoparticles. 05_Plasmonic_Heating.pdf	Mie/dipole optics, finite-beam photon budget, transient Au/water heat. “Optical theorem, Mie theory and spectral results”; “Extreme regimes, admissible candidates and formal continuation.”
6	Frozen Geometry AuNP–Bubble Thermal Coupling. 06_AuNP_Bubble_Coupling.pdf	Finite-Au stationary axisymmetric transport and its applicability limits. “Resolved thermal problem and the signed interface”; “Kinematics, event resolution and applicability”; “Local energy, multiple particles and the reusable channel.”
7	Integrated Laser–AuNP–Bubble Model: Self-Consistent Dynamics and Peak Gas Temperature. 07_Integrated_Model.pdf	Moving conservative water, paired laser peaks, feedback and formal extrema. “One signed interface, gas energy and mechanical feedback”; “Paired temperatures and timing sensitivity”; “Coefficient domains, formal extrema and physical interpretation.”

The separate Mini-Research 8 Technical Report gives the bounded new comparison, caloric operations and energy audits underlying this synthesis. Its numerical package contains data, model code and reproducible checks. No supporting report is needed to identify the principal assumptions and scientific conclusion stated above.